PROJECT REPORT

Ecommerce Transaction Analysis

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1. Introduction

The rapid growth of e-commerce platforms has generated large volumes of transactional data. Businesses rely on data analytics to understand customer behavior, optimize product offerings, and improve revenue performance.

This project analyzes an ecommerce transaction dataset containing **50,000 transactions across multiple countries and product categories**. The goal is to extract actionable insights related to:

- Sales performance
- Customer purchasing behavior
- Geographic distribution of revenue
- Product category performance
- Payment method preferences

The analysis pipeline includes:

- Data cleaning using **Python (Pandas)**
- Analytical querying using **SQL**
- Interactive visualization using **Power BI**

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2. Dataset Overview

The dataset contains ecommerce transaction records with the following attributes:

Column	Description
Transaction_ID	Unique identifier for each transaction
User_Name	Customer name
Age	Customer age
Country	Country where transaction occurred
Product_Category	Category of product purchased
Purchase_Amout	Transaction value
Payment_Method	Payment method used
Transaction_Date	Date of transaction

Dataset size:

- **Total Transactions:** 50,000
- **Countries:** 10
- **Product Categories:** 8

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3. Data Cleaning and Preparation

Data cleaning was performed using **Python and Pandas** to ensure data consistency and prepare it for analysis.

Key steps included:

- Loading the raw dataset
- Converting date columns into datetime format
- Removing duplicate records

- Validating purchase amounts
- Creating additional features for analysis

Feature Engineering

Additional columns were created:

- Year
- Month
- Day
- Day_of_Week

These features enable time-based analysis and trend detection.

Age Group Segmentation

Customers were segmented into demographic groups:

Age Range	Group
18–25	Young Adults
26–35	Early Professionals
36–45	Mid Career
46–60	Experienced Professionals
60+	Seniors

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4. Database Integration

The cleaned dataset was loaded into a **MySQL database** to perform SQL-based analytics.

Database: ecommerce

Table: transactions

Python's **SQLAlchemy** library was used to transfer the cleaned dataset from Pandas to MySQL.

Example operation: dataset.to_sql("transactions", engine)

This allowed efficient querying and aggregation using SQL.

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5. SQL Analysis

SQL queries were written to answer key business questions.

Key Business Questions

1. Which product categories generate the highest revenue?
2. Which countries generate the most sales?
3. What are the monthly sales trends?
4. Which payment methods generate the highest revenue?
5. What is the average purchase amount per country?
6. Which product categories are most popular in each country?
7. What age group spends the most money?
8. Who are the top customers by spending?
9. Which day of the week has the highest sales?
10. What percentage of revenue comes from each product category?

Advanced SQL techniques used:

- Aggregations (`SUM`, `AVG`, `COUNT`)

- Window functions (`RANK`)
- Subqueries
- Customer lifetime value analysis
- Revenue contribution analysis

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6. Power BI Dashboard

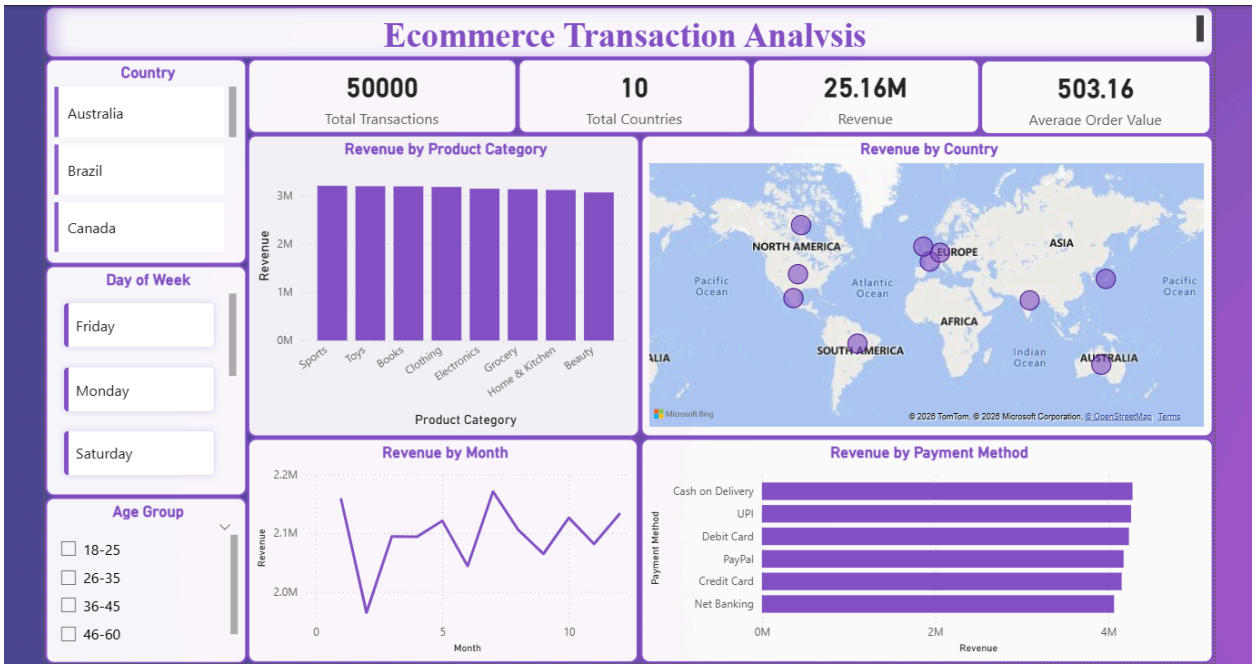
An interactive dashboard was developed using **Power BI** to visualize insights and enable dynamic exploration of the data.

Dashboard features include:

- KPI summary metrics
- Product category revenue analysis
- Geographic sales distribution
- Monthly revenue trends
- Payment method revenue comparison
- Interactive filters for segmentation

Dashboard Screenshot

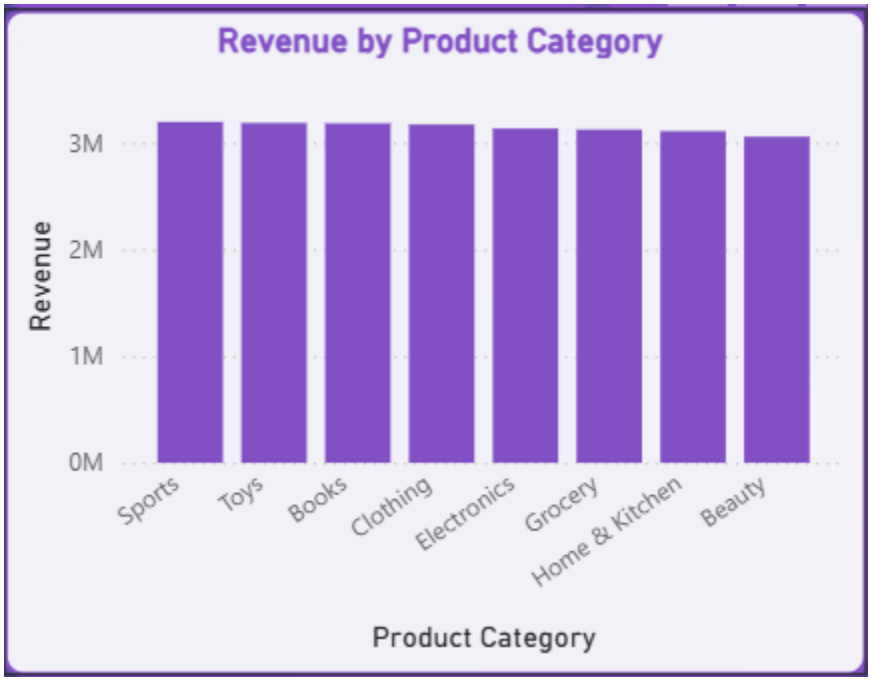
Add screenshot of full Power BI dashboard



7. Key Visualizations

Revenue by Product Category

This visualization shows which product categories contribute the most to total revenue.

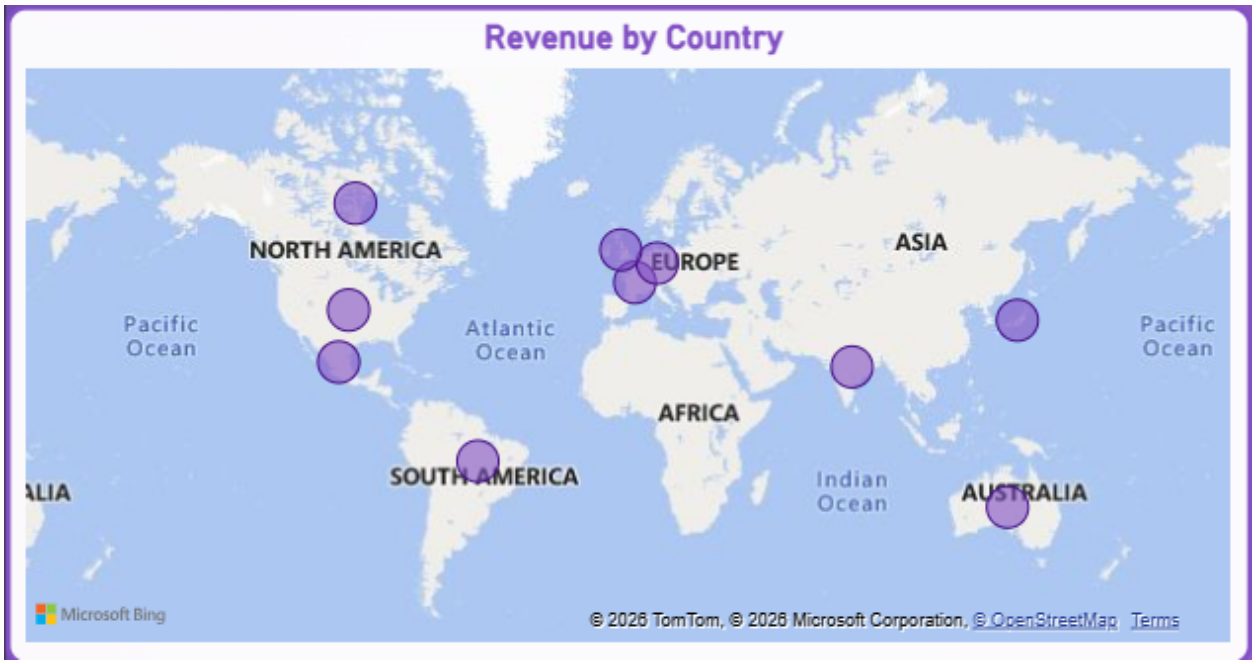


Insights:

- Some product categories significantly outperform others.
- High-performing categories can be prioritized in marketing strategies.

Revenue by Country

A geographic map visualization displays revenue distribution across countries.

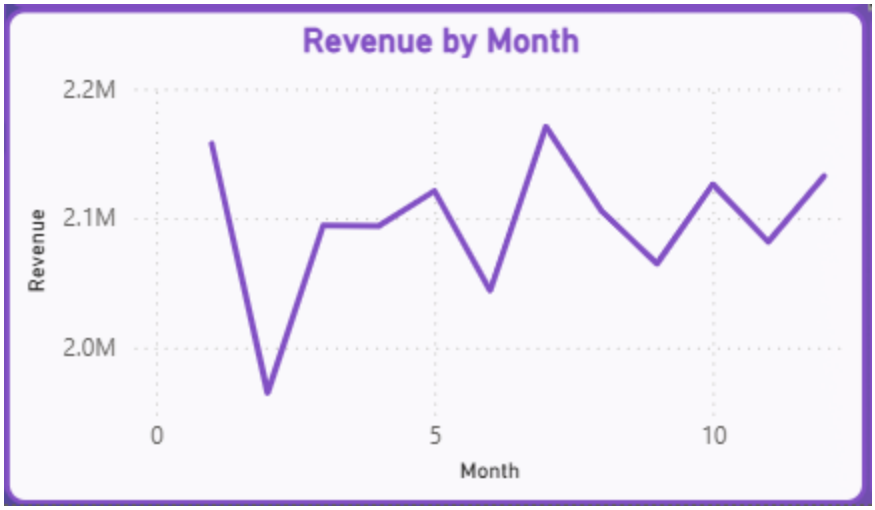


Insights:

- Certain regions generate significantly higher revenue.
- Geographic insights can guide regional expansion strategies.

Monthly Revenue Trend

A line chart tracks monthly sales performance.

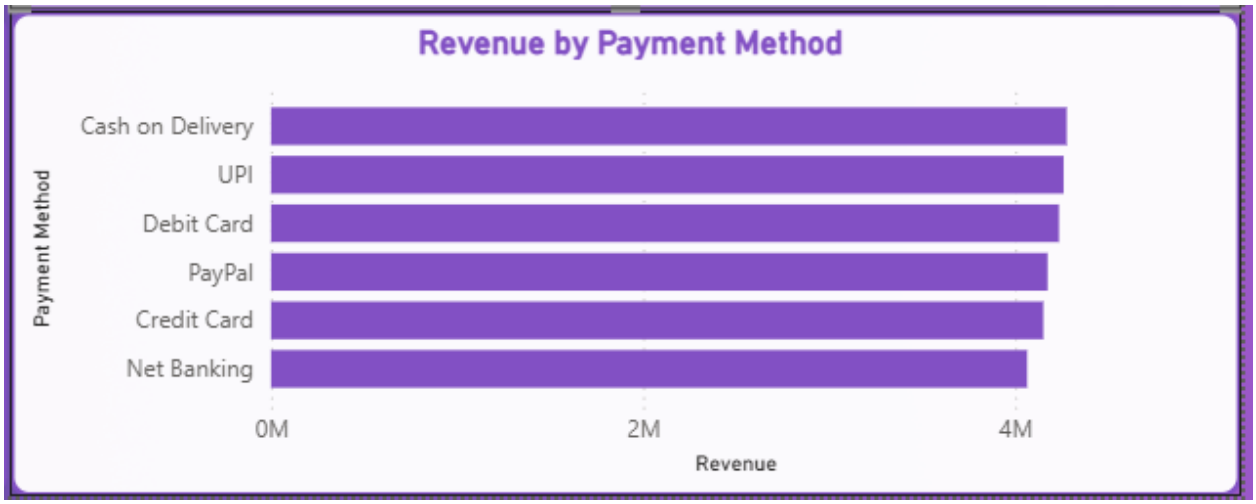


Insights:

- Revenue remains relatively stable across months.
- Identifying seasonal spikes can help plan promotional campaigns.

Revenue by Payment Method

This visualization compares revenue generated through different payment methods.



Insights:

- Digital payments dominate transactions.

- Businesses should prioritize optimized online payment systems.

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8. Key Insights

Several important business insights were discovered:

1. Certain product categories contribute the majority of total revenue.
2. Revenue distribution varies significantly across countries.
3. Digital payment methods such as UPI and cards dominate transactions.
4. Monthly revenue trends remain relatively consistent.
5. Specific customer segments generate higher spending.

These insights can help businesses optimize:

- marketing strategies
- product assortment
- regional expansion
- payment infrastructure

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9. Conclusion

This project demonstrates a complete data analytics workflow:

- Data cleaning using Python
- Data analysis using SQL

- Data visualization using Power BI

By combining these tools, meaningful insights were extracted from transactional data.

The dashboard provides a clear overview of business performance and enables stakeholders to make data-driven decisions.

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10. Future Improvements

Future improvements could include:

- Customer segmentation using clustering
- Sales forecasting using time-series models
- Recommendation systems for personalized product suggestions
- Advanced cohort analysis